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Hardware

UNO R3



Tutorials

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Analog Input

Use a potentiometer to control the blinking of an LED.

Last revision • 10/02/2024

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Analog Input

Use a potentiometer to control the blinking of an LED.

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In this example we use a variable resistor (a potentiometer or a photoresistor), we read its value using one analog input of an Arduino board and we change the blink rate of the built-in LED accordingly. The resistor's analog value is read as a voltage because this is how the analog inputs work.

Hardware Required

Arduino Board

Potentiometer **or**

10K ohm
photoresistor and 10K
ohm resistor

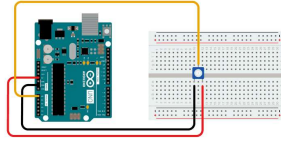
built-in LED on pin 13
or

220 ohm resistor and
red LED

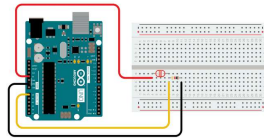
Circuit

With a potentiometer

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With a photoresistor



Connect three wires to the Arduino board. The first goes to ground from one of the outer pins of the potentiometer. The second goes from 5 volts to the other outer pin of the potentiometer. The third goes from analog input 0 to the middle pin of the potentiometer.

For this example, it is possible to use the board's built in LED attached to pin 13. To use an additional LED, attach its longer leg (the positive leg, or anode), to digital pin 13 in series with the 220 ohm resistor, and it's shorter leg (the negative leg, or cathode) to the ground (GND) pin next to pin 13.

The circuit based on a photoresistor uses a resistor divider to allow the high impedance Analog input to measure the voltage. These inputs do not draw almost any current, therefore by

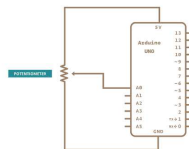
of a resistor connected to 5V is always 5V, regardless the resistor's value. To get a voltage proportional to the photoresistor value, a resistor divider is necessary. This circuit uses a variable resistor, a fixed resistor and the measurement point is in the middle of the resistors. The voltage measured (V_{out}) follows this formula:

$$V_{out} = V_{in} * (R_2 / (R_1 + R_2))$$

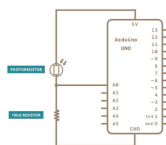
where V_{in} is 5V, R_2 is 10k ohm and R_1 is the photoresistor value that ranges from 1M ohm in darkness to 10k ohm in daylight (10 lumen) and less than 1k ohm in bright light or sunlight (>100 lumen).

Schematic

Potentiometer



Photoresistor



Code

`sensorPin` is set to to analog pin 0, where your potentiometer is attached, and `ledPin` is set to digital pin 13. You'll also create another variable, `sensorValue` to store the values read from your sensor.

The `analogRead()` command converts the input voltage range, 0 to 5 volts, to a digital value between 0 and 1023. This is done by a circuit inside the microcontroller called an *analog-to-digital converter* or *ADC*.

By turning the shaft of the potentiometer, you change the amount of resistance on either side of the center pin (or wiper) of the potentiometer. This changes the relative resistances between the center pin and the two outside pins, giving you a different voltage at the analog input. When the shaft is turned all the way in one direction, there is no resistance between the center pin and the pin connected to ground. The voltage at the center pin then is 0 volts, and `analogRead()` returns 0. When the shaft is turned all the way in the other direction, there is no resistance between the center pin and the pin

then is 5 volts, and

`analogRead()` returns

1023. In between,

`analogRead()` returns a

number between 0 and 1023

that is proportional to the

amount of voltage being

applied to the pin.

That value, stored in

`sensorValue`, is used to set

a `delay()` for your blink

cycle. The higher the value,

the longer the cycle, the

smaller the value, the

shorter the cycle. The value

is read at the beginning of

the cycle, therefore the

on/off time is always equal.

```
1  /*
2     Analog
3     Input
4
5     Demonstrates
6     analog
7     input by
8     reading an
9     analog
10    sensor on
11    analog pin
12    0 and
13    turning
14    on and off
15    a light
16    emitting
17    diode(LED)
18    connected
19    to digital
20    pin 13.
21
22    The
23    amount of
24    time the
25    LED will be
26    on and off
```

Learn more

You can find more basic tutorials in the [built-in examples](#) section.

You can also explore the [language reference](#), a detailed collection of the Arduino programming language.

Last revision 2015/07/28 by SM

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